

Garv Arora

Robotics & Systems Engineer | Autonomous Systems · ROS2 · C/C++ · Python · Embedded RTOS · CV · AI Infra
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Education

Vellore Institute of Technology (VIT)

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

Vellore, India

08/2023 – 07/2027

Gems International School (CBSE)

Class XII: 76.4% | Class X: 89.3%

Gurugram, India

11/2016 – 04/2023

Patents & Key Honors

Patent Published (App No: IN202641072249 A1): Autonomous Radar-Guided Survivor Detection and Navigation System — Fused 24GHz FMCW mmWave radar, ultrasonic array, and IMU dead-reckoning on dual-core ESP32; formulated confidence grid decay $C = \max(0, C - \beta\Delta t)$ for survivor breathing detection under rubble. (06/2026)

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate).

International Rover Challenge (IRC): Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT).

Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner | Oracle Cloud Data Science & GenAI Professional.

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS

Work Experience

SEDS India (SEDS VIT)

Vellore, India

Research & Development Lead

01/2026 – Present

- Promoted to R&D Lead directing 2 space robotics competition teams (50+ engineers) across Team Vyadh (IRC) and Team Ardra (ISDC); conducted architecture reviews and hardware debugging, resolving 10+ critical pre-competition issues.

Autonomous Systems Developer

04/2024 – 01/2026

- Architected ROS2 Nav2 & RTAB-Map 3D Visual SLAM pipeline on Intel RealSense D455, generating real-time elevation maps and slashing manual teleoperation time by 70% during autonomous competition traversals.
- Slashed CV pipeline latency by ~60% and boosted throughput 4x using CUDA GPU kernels, YOLOv8, ArUco/AprilTag tracking, and Linux V4L2 zero-copy buffer tuning across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware onto ESP32 microcontrollers for distributed sensor polling, fusing wheel encoders and 6-DOF IMU with an Extended Kalman Filter (EKF) to cut odometry drift by ~40% (sub-10 cm positioning accuracy).
- Shipped PyQt5 mission dashboard with real-time telemetry visualization, parameter tuning, and FSM-based autonomous mission execution.

Samsung PRISM

Remote

Data Intelligence Agent Developer (Internship)

08/2025 – Present

- Developed multi-agent LangGraph data engine (FastAPI, Redis, PostgreSQL, Qdrant) automating discovery, screening, and 20-stage analysis of HuggingFace/Kaggle corpora; integrated spaCy NLP, Presidio PII anonymization, and MemorySaver HITL pause.

LG Soft India (LGSi) – HS/ES India Lab

Bengaluru, India

Living Solution Control Development Intern

06/2026 – 06/2026

- Engineered C/C++ embedded control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; work certified as “Technically Competent” by LGSi HR/Engineering management.

Wissen Baum Engineering Solutions LLP

Pune, India

Software Automation Intern

05/2025 – 07/2025

- Built Python BDD test automation framework (Gherkin/Behave, pytest, Playwright), parallelizing GitLab CI jobs to cut pipeline execution time from ~10m to ~6m (40% speedup) and reducing manual validation effort by 80%+.

Projects

HayaiOS – Preemptive Bare-Metal RTOS Kernel | C, ARM Assembly, ARM Cortex-M4

01/2026 – 03/2026

- Implemented preemptive scheduler (1 ms tick) with context switching in <50 ARM assembly instructions, mutex/semaphore IPC primitives, and a hardware abstraction layer (HAL) isolating register-level I/O on ARM Cortex-M4.

Captivity CLI | Rust, Python, Linux D-Bus, systemd, Socket IPC | pypi.org/project/captivity-cli

11/2025 – 01/2026

- Built autonomous captive portal WiFi login daemon featuring <50ms HTTP 204 probing, systemd background integration, D-Bus NetworkManager monitoring, keyring credential encryption, and dual Python-Rust TCP socket IPC.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

08/2025 – 10/2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and 6-DOF IMU data via complementary filter, reducing tracking failures by ~50% across fast-motion 6-DOF trajectories.

teleop-cursor | ROS 2, Python, rclpy, geometry_msgs/Twist | pypi.org/project/teleop-cursor

02/2026 – 04/2026

- Developed zero-hardware mouse cursor teleoperation node for ROS 2, converting screen displacement vectors into smooth linear/angular velocity commands (*geometry_msgs/Twist*) at 10 Hz callback frequency for TurtleBot3/Gazebo.